Chapter 1

Introduction

1.1 Ordinary Differential Equations (ODE)

Solving an ordinary differential equation of order k is the task of finding a function $y : \mathcal{D} \subset \mathbb{R} \to \mathbb{R}$, such that $y(x)$ satisfies

$$G(y^{(k)}, y^{(k-1)}, \ldots, y, x) = 0, \qquad \forall x \in \mathcal{D}, \tag{1.1}$$

where $G : \mathbb{R}^{k+1} \times \mathcal{D} \to \mathbb{R}$ is a given function. The notation $y^{(k)}$ denotes the kth derivative of the function y. (See appendix A.2 for general notation on functions and derivatives.)

Here we concentrate on *explicit* ODE of the form

$$y^{(k)} = F(y^{(k-1)}, \ldots, y, x), \tag{1.2}$$

where $F : \mathbb{R}^k \times \mathcal{D} \to \mathbb{R}$ is a given function. An ODE of the form (1.2) is called explicit because one can isolate the highest derivative on one side. This situation arises often when modeling real-life problems, but certainly not always.

In addition to functional relations between the derivatives, differential equations typically involve *additional constraints*, e.g. initial conditions

$$y(x_0) = c_0, \quad \ldots \quad , y^{(k-1)}(x_0) = c_{k-1}. \tag{1.3}$$

Such additional conditions are crucial for the existence of a unique solution. Before we quote some general theorems on the existence and uniqueness of solutions to ODE, we consider a simple example:

Example 1.1 Consider a simple pendulum under the influence of gravity. Figure 1.1 gives a description of the problem. We seek to characterize the pendulum's motion in terms of the angle $\theta(t)$. Taking the moment balance about the pivot yields

$$ml^2\theta'' = -mgl\sin(\theta),$$

where ml^2 is the moment of inertia, and g is the gravitational acceleration. We obtain an explicit (nonlinear) ODE of order 2:

$$\theta'' = \underbrace{-\frac{g}{l}\sin(\theta)}_{F(\theta)}.$$

For small angles, we have $\sin\theta \approx \theta$, and thus

$$\theta'' + \frac{g}{l}\theta \quad = \quad 0. \tag{1.4}$$

We note that any function

$$\theta(t) = A \sin\left(\sqrt{\frac{g}{l}}t\right) + B \cos\left(\sqrt{\frac{g}{l}}t\right)$$

where A and B are arbitrary constants, satisfies (1.4). Apparently specifying only the ODE (1.4) is not enough to completely determine the solution.

Imagine now that we hold the pendulum fixed at some angle θ_0, and at time $t = 0$ we simply let it go. Mathematically this can be described as

$$\theta(t=0) = \theta_0, \tag{1.5}$$

$$\theta'(t=0) = 0. \tag{1.6}$$

This is an *initial condition* of the form (1.3). We can readily verify that

$$\theta(t) = \theta_0 \cos\left(\sqrt{\frac{g}{l}}t\right)$$

solves the ODE with given initial conditions (1.5), (1.6). In this case, we can write the solution in closed form. (This is not at all guaranteed!)

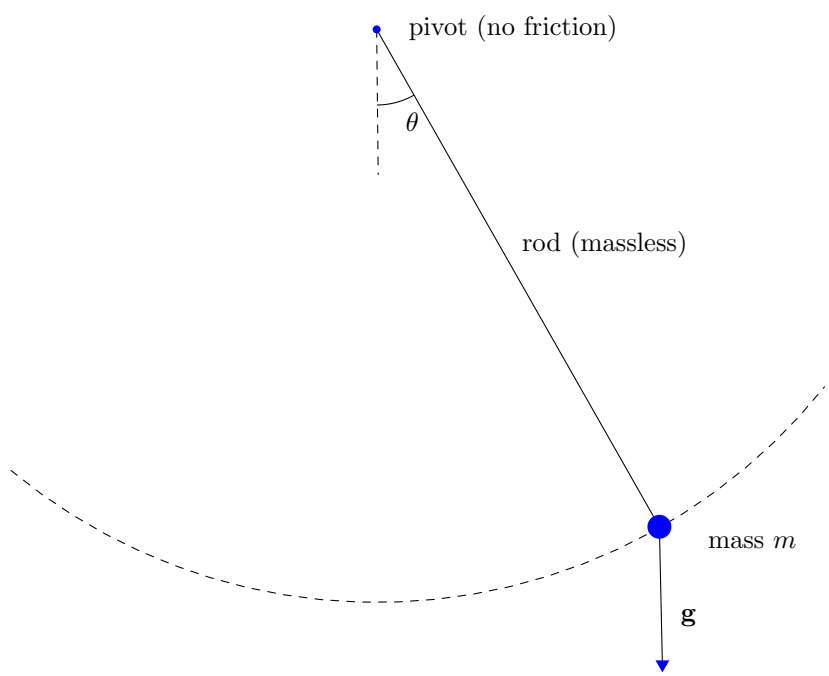

Figure 1.1: A simple gravity pendulum. The moment about the pivot point exerted by gravity is given by $M_p = -F_T l$, where l is the length of the rod, and $F_T = mg \sin\theta$. If we treat the mass m as a point mass, this causes angular acceleration via $I\theta'' = M_p$, where $I = ml^2$ is the moment of inertia of a point mass. (The prime superscripts indicate differentiation with respect to time. We take the angle positive in a counterclockwise sense.)

1.2 Existence and Uniqueness of Solutions to ODE

Consider an ODE of order k. We can transform a kth order scalar ODE of the form

$$y^{(k)} = F(y^{(k-1)}, \dots, y, x)$$

into a *system* of first order ODE. Let $\mathbf{y} = (y_1, y_2, \dots, y_k)^T$ be a vector-valued function, where

$$\begin{aligned} y_1 &= y, \\ y_2 &= y', \\ &\vdots \\ y_k &= y^{(k-1)}. \end{aligned}$$

Define differentiation of $\mathbf{y}$ component wise. Then

$$\mathbf{y}' = \begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_k' \end{pmatrix} = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ F(y_k, y_{k-1}, \dots y_1, x) \end{pmatrix} =: \mathbf{F}(\mathbf{y}, x). \tag{1.7}$$

We can thus transform a kth order scalar equation of the kind (1.2) into a first order system of ODE. Fortunately, there is a general theorem regarding existence and uniqueness of solutions to such systems. The only technical tool we need is *Lipschitz continuity*:

Definition 1.1 (Global Lipschitz Continuity) *For $a, b \in \mathbb{R}$, and $a < b$, let $S := \{(\mathbf{y}, x) : x \in [a, b], \mathbf{y} \in \mathbb{R}^k\}$. Consider a function $\mathbf{F} : S \to \mathbb{R}^k$. We say $\mathbf{F}(\mathbf{y}, x)$ satisfies a* Lipschitz condition *on $\mathbf{y}$ in S, if there exists a positive constant L, such that*

$$\|F(\mathbf{y}_\alpha, x) - F(\mathbf{y}_\beta, x)\| \leq L \|\mathbf{y}_\alpha - \mathbf{y}_\beta\| \tag{1.8}$$

for all $x \in [a, b]$, and $\mathbf{y}_\alpha$, $\mathbf{y}_\beta \in \mathbb{R}^k$.

Theorem 1.1 (Picard-Lindelöf) *Let $\mathbf{F}(\mathbf{y}, x)$ and S be as in Definition 1.1. In particular, let $\mathbf{F}$ satisfy the Lipschitz condition* (1.8)*, and be continuous in x. Then for every $x_0 \in [a, b]$ there is a unique continuously differentiable function $\mathbf{y} : [a, b] \to \mathbb{R}^k$, which satisfies*

$$\begin{aligned} \mathbf{y}' &= F(\mathbf{y}, x), \qquad x \in [a, b], \\ \mathbf{y}(x_0) &= \mathbf{y}_0. \end{aligned}$$

The theorem requires global Lipschitz continuity, and gives likewise a global solution. This is the best-case scenario. If $\mathbf{F}$ is Lipschitz continuous only in a *subset* of $U \subset S$, then we are still guaranteed a *local solution*. This means a solution exists in a neighborhood of the initial conditions. While it is often a nontrivial problem to ascertain more information on how far away from the initial condition a solution exists, it should still be appreciated that the existence of a (local) unique solution to *any* ODE that can be written as a first order system, and which satisfies a Lipschitz condition, is quite remarkable. In this course, however, we mainly deal with partial differential equations (PDE) to be defined below. Unfortunately there is no corresponding result for PDE.

1.3 Partial Differential Equations

We define a kth order *partial differential equation* (PDE) as an expression of the form

$$F(D^k u(x), \ldots, Du(x), u(x), x) = 0, \qquad x \in \mathcal{D} \subset \mathbb{R}^d, \quad d \in \mathbb{N}.$$

Here F is given, and $u : \mathcal{D} \to \mathbb{R}$ is the unknown. (Recall from appendix A.2 that the notation D^k denotes a tuple containing all partial derivatives of order k.) In general u is a function of several independent variables, and hence derivatives are partial derivatives. A PDE thus constitutes a relation between partial derivatives of a function. As mentioned above, there is no general solution theorem, similar in scope to Theorem 1.1, for PDE. Perhaps it is best to again start with a few examples.

1.3.1 The Heat Equation

Consider a very long thin wire of length L. We assume the wire is thermally insulated (except at the endpoints), and that there are no heat sources, or work done. We would like to know the

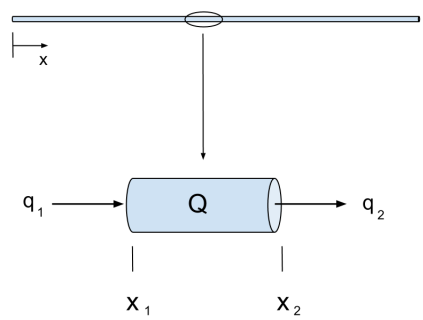

Figure 1.2: Analysis of the heat conduction problem via control volumes.

temperature distribution inside the wire, given

- the initial temperature distribution $\phi(x)$ at time $t = 0$,
- prescribed temperature at the endpoints, say $h_1(t)$ at $x = 0$, and $h_2(t)$ at $x = L$.

We derive the governing equation for this problem, proceeding in an ad hoc manner, using only basic notions from thermodynamics. Since the wire is very long and thin, we neglect temperature variations in the cross-section, and treat this as a (spatially) one-dimensional problem. We monitor the rate of change of internal energy in a control volume between x_1 and x_2. For many materials, and a wide range of temperatures, one can assume that the rate-of-change of the internal energy is proportional to the rate-of-change of the temperature. The constant of proportionality is the *heat capacity*:

$$Q(x,t) = Q_0 + c(x)\,T(x,t), \tag{1.9}$$

where

- Q is the *specific* internal energy, i.e. energy per unit mass ($[Q] = J/kg$);
- c is the specific heat capacity per unit mass ($[c] = J/(kgK)$);
- T is the temperature ($[T] = K$).

The quantity Q_0 is a reference energy for $T = 0$. It has no consequences for the derivation to follow, and we may safely set it to zero. Let A be the area of the wire's cross-section. We assume that A does not change with x. Consider a small volume element Adx contained in our control volume between x_1 and x_2. The mass contained in that small volume element is

$$dm = \rho(x)Adx, \tag{1.10}$$

where ρ is the density of the material. The total internal energy inside the control volume between x_1 and x_2 at time t is obtained by integrating the specific internal energy over the mass contained in the control volume. By (1.10) we have

$$E(t; x_1, x_2) = A \int_{x_1}^{x_2} \rho(x)Q(x,t)dx. \tag{1.11}$$

As we have assumed perfect insulation there is no loss of heat through temperature exchange with the surroundings. In the absence of a heat source, the rate-of-change of internal energy is then due solely to *heat flux* across the boundaries at x_1 and x_2 of the control volume,

$$\frac{d}{dt}E = A(q(x_2) - q(x_1)), \tag{1.12}$$

where q is the heat flux ($[q] = J/(m^2 s) = W/m^2$), i.e. the amount of energy lost or gained per unit time, and per unit area. Figure 1.2 shows an illustration. As we have assumed a one-dimensional setting we stipulate that the heat flux is constant over the cross-section of the wire. Fourier's law says that the heat flux follows the negative temperature gradient:

$$q(x) = -\kappa(x)\frac{\partial T}{\partial x}, \tag{1.13}$$

where κ is the conductivity of the material ($[\kappa] = W/(mK)$). Substituting (1.13) into (1.12), and noting (1.11), we obtain

$$\begin{aligned} \frac{d}{dt}\int_{x_1}^{x_2} \rho(x)Q(x,t)dx &= \kappa(x_2)\frac{\partial T}{\partial x}(x_2) - \kappa(x_1)\frac{\partial T}{\partial x}(x_1) \\ &= \int_{x_1}^{x_2} \frac{\partial}{\partial x}\left(\kappa(x)\frac{\partial T}{\partial x}\right) dx. \end{aligned} \tag{1.14}$$

Now substitute (1.9) into (1.14) to obtain

$$\int_{x_1}^{x_2} \left(\rho(x)c(x)\frac{\partial T}{\partial t} - \frac{\partial}{\partial x}\left(\kappa(x)\frac{\partial T}{\partial x}\right)\right) dx = 0.$$

The control interval is arbitrary. Hence the integrand must vanish, so that we finally obtain:

$$\rho(x)c(x)\frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(\kappa(x)\frac{\partial T}{\partial x}\right).$$

Let us assume constant coefficients. Defining $a = \kappa/(\rho c)$ we have a mathematical formulation of our problem:

$$\begin{aligned} \frac{\partial T}{\partial t} &= a\frac{\partial^2 T}{\partial x^2}, & (x,t) \in (0,L) \times (0,\infty) \\ T(x,0) &= \phi(x), & x \in [0,L] \\ T(0,t) &= h_1(t), & t > 0 \\ T(L,t) &= h_2(t), & t > 0 \end{aligned} \tag{1.15}$$

To give this a more physical meaning, let us assume that we have a means of fixing the temperature at both end points to the same temperature T_0. Without loss of generality, we can

set $T_0 = 0$.[1] So let $h_1 = h_2 = 0$, and $\phi(x)$ be continuous, such that it fulfills the boundary conditions at $x = 0$, and $x = L$. Following Appendix B.2, a general solution is given by:

$$T(x,t) = \sum_{n=1}^{\infty} c_n \sin(\lambda_n x)\, e^{-a\lambda_n^2 t} \tag{1.16}$$

where

$$\lambda_n = \frac{n\pi}{L}, \qquad c_n = \frac{2}{L}\int_0^L \sin(\lambda_n x)\,\phi(x)\,dx.$$

Under the assumptions made for the initial conditions, the Solution (1.16)

- possesses *infinitely* many derivatives, even when the initial conditions do not,
- depends on $\phi(y)$ for all $y \in [0, L]$, at any point (x, t), $x \in (0, L)$, $t > 0$.

The remarkable smoothness of solutions to the heat equation comes from the fact that, from a molecular perspective, heat conduction is a diffusive process. (It is basically diffusion of molecular kinetic energy.) It is a hallmark of diffusive processes that features of the solution are "smeared out". This can also be observed in other diffusive processes, such as species diffusion by Fick's law, which often follow very similar governing equations. It should also be observed that this diffusive smoothing happens infinitely fast: The solution becomes smooth instantly, and also the solution at any point in the computational domain is affected by the all of the initial conditions. Apparently then, the heat equation admits infinitely fast speed of propagation. This is certainly a consequence of the idealized modeling, but also a hallmark of equations that model diffusive phenomena.

1.3.2 The Laplace Equation

A derivation similar to the one undertaken in the previous section leads to the 2D heat equation: Let $(x, y) \in \Omega \subset \mathbb{R}^2$, where Ω is some open and bounded domain with boundary $\partial\Omega$. The heat equation with prescribed initial temperature distribution in Ω, and prescribed temperature on the boundary $\partial\Omega$ reads

$$\begin{aligned} \frac{\partial T}{\partial t} &= a\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) && (x,y,t) \in \Omega \times (0,\infty), a > 0, \\ T(x,y,0) &= g(x,y) && (x,y) \in \Omega, \\ T(x,y,t) &= h(x,y,t) && (x,y) \in \partial\Omega,\ t > 0. \end{aligned} \tag{1.17}$$

We can think of the domain Ω as a thin plate. If $h \neq h(t)$, we may only be interested in a stationary temperature distribution, which is achieved after some time, when the system has reached an equilibrium. Putting the time derivative to zero in (1.17), we postulate that such a stationary temperature distribution obeys the *Laplace Equation*:

$$\begin{aligned} \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} &= 0 && (x,y) \in \Omega, \\ T(x,y) &= h(x,y) && (x,y) \in \partial\Omega. \end{aligned}$$

1.3.3 The Vibrating String Problem

Let a flexible, elastic thread undergo small transverse motion (cf. Fig. 1.3). We "cut" the string at x_1 and x_2 and assume that

- There is motion only in y direction;

[1] It is clear from the heat equation that a constant offset does not matter. Hence we can define the zero-point of our temperature scale anywhere we like.

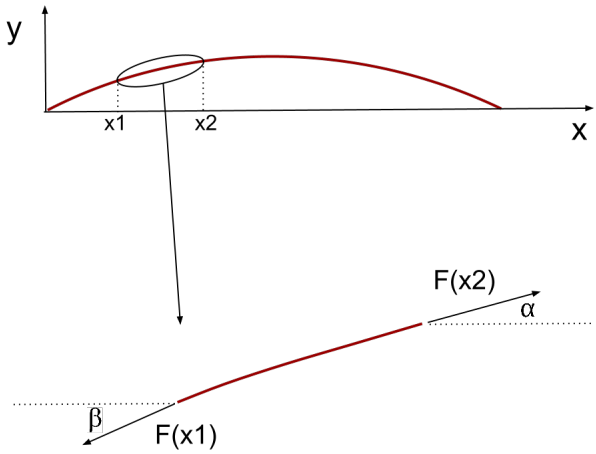

Figure 1.3: A piece "cut" out from a vibrating string, and the internal forces acting on it.

- Forces are in equilibrium in x-direction;
- External forces (gravity!) are negligible;
- There are no shear stresses (forces act in the direction of the string).

Postulating equilibrium in x direction yields

$$0 = F(x_1)\cos(\alpha) - F(x_2)\cos(\beta).$$

Assuming small angles, i.e. small deflections, we set $\cos(\alpha) \approx \cos(\beta) \approx 1$. Hence

$$F(x_1) = F(x_2) = F.$$

Consider for some x between x_1 and x_2 the mass contained in an infinitesimal volume element:

$$dm = A\rho(x)dx,$$

where ρ is the density, and A is the cross-section (assumed to be constant). According to Newton's law the acceleration of mass between x_1 and x_2 must be proportional to net force in y-direction

$$A\frac{d^2}{dt^2}\int_{x_1}^{x_2} \rho(x)y\,dx = F\sin\alpha - F\sin\beta.$$

We use $\sin(\alpha) \approx \alpha \approx \tan(\alpha)$, and note that $\tan(\alpha)$ can be characterized as

$$\tan\alpha = \lim_{h\to 0}\frac{y(x_2+h) - y(x_2)}{h} = \frac{\partial y}{\partial x}(x_2).$$

Similarly, $\tan\beta = y_x(x_1)$. Hence we have

$$\begin{aligned} A\int_{x_1}^{x_2} \rho(x)\frac{\partial^2 y}{\partial t^2}\,dx &= F\left(\frac{\partial y}{\partial x}(x_2) - \frac{\partial y}{\partial x}(x_1)\right) \\ &= F\int_{x_1}^{x_2}\frac{\partial^2 y}{\partial x^2}\,dx. \end{aligned}$$

Since the control volume is arbitrary, we can write

$$\frac{\partial^2 y}{\partial t^2} - c^2\frac{\partial^2 y}{\partial x^2} = 0,$$

where $c^2 = F/(A\rho)$. (We assume $\rho = const.$) For convenience, we assume that the string is infinite:

$$\frac{\partial^2 y}{\partial t^2} - c^2 \frac{\partial^2 y}{\partial x^2} = 0, \qquad (x,t) \in \mathbb{R} \times (0,\infty).$$

At $t = 0$, we have initial conditions

$$\begin{aligned} y(x,0) &= \phi(x), \qquad x \in \mathbb{R}, \\ \frac{\partial y}{\partial t}(x,0) &= \psi(x), \qquad x \in \mathbb{R}. \end{aligned}$$

We set $\psi(x) \equiv 0$. This corresponds to a situation where we move the string according to a prescribed function ϕ in y-direction, and at time $t = 0$ we simply let it go. Following appendix B.3 we have the solution

$$y(x,t) = \frac{1}{2}\left(\phi(x+ct) + \phi(x-ct)\right).$$

Especially note the following properties, and contrast them with the heat equation in section 1.3.1:

- Smoothness of y is *inherited* from ϕ. This means, y possesses as many continuous derivatives as ϕ does;
- The solution at (x,t) depends only on $\phi(y)$ for $y \in [x-ct, x+ct]$.

As opposed to the heat equation, the wave equation admits solutions that model *finite* speed of propagation. For example, if you take initial conditions as a sharp pulse centered at some location on the real line, then that pulse is simply propagated to the left and right with speed c. (This appears like traveling waves, hence the name of the equation.) The wave equation is thus a model for transport (or: convective) processes.

1.4 Course Content

In this brief introduction we have seen three different PDE. These PDE are representative of important classes of PDE. The heat equation is of the type

$$u_t - u_{xx} = 0.$$

This is a *parabolic* PDE. The wave equation is of the type

$$u_{tt} - u_{xx} = 0.$$

This is a *hyperbolic* PDE. The stationary heat equation is of the type

$$u_{xx} + u_{yy} = 0.$$

This is an *elliptic* PDE. The particular choice of label that we have attached to these equations has to do with some similarities to the classification of conic sections. A short overview is given in Appendix B.1. Different types of PDE are quite distinct from one another, even if they look somewhat similar. For example, the heat equation and the wave equation have very different behavior. We have already noted this in the derivation above. Moreover, in our previous derivation, the Laplace equation (i.e. an elliptic equation) models a stationary process, while the other two types of equations have modeled time-dependent processes.

In this course we discuss numerical methods for all of the above types. In particular, we cover three main topics:

Finite Difference Methods

- Mostly for parabolic PDE

Finite Volume Methods

- Mostly for hyperbolic PDE

Finite Element Methods

- First for elliptic PDE
- Then for other (mixed) types

Pairing numerical schemes and types of equations in this manner does not imply that such pairing is exclusive. Any of the above-mentioned numerical methods can be used with any of the types of PDE. However, it seems advantageous to start the course with a combination of parabolic equations and finite difference methods, since both are relatively benign: Parabolic equations have very smooth solutions, finite difference methods are straightforward to derive. It is a good framework to illustrate some very basic principles of numerical approximation.

Hyperbolic problems, on the other hand, often lead to solutions that are less well behaved. This is true in particular for nonlinear equations. For such equations, finite-volume methods are the most popular type of numerical method.

Finite element methods are perhaps the overall most popular numerical method for complex PDE problems. The theory is best developed for elliptic equations, and we shall start there. However, we also formulate finite-element methods for other types of equations in the end.